Sakshi Gupte

Data Scientist

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SUMMARY

Data Scientist with 4+ years of experience in data extraction, modeling, wrangling and machine learning to drive actionable insights. Skilled in advanced algorithms like Regression, XGBoost, Random Forests, SVM, and NLP, with a focus on predictive modeling. Proficient in Python (NumPy, Pandas, SciPy) and integrating AI technologies such as LLMs for enhanced decision-making. Experienced in data visualization using Tableau, Power BI and Excel with expertise in cloud platforms (AWS, GCP) and database management (MySQL, SQL Server, MongoDB). Strong in data cleaning, modeling, and stakeholder collaboration to deliver impactful results.

SKILLS

Methodologies: SDLC, Agile, Waterfall
Language/IDEs/Statistical Methods: Python, R, SQL, SAS, Java, Visual Studio Code, PyCharm, Jupyter Notebook, Hypothetical Testing, ANOVA
Machine Learning: Regression analysis, Bayesian Method, Decision Tree, Random Forests, Support Vector Machine, Neural Network, Sentiment Analysis, K-Means Clustering, KNN, Classification, SVM, Naive Bayes, Natural Language Processing (NLP), LLM, CNN, XGBoost, Deep Learning
Packages: NumPy, Pandas, Matplotlib, SciPy, ggplot2, Scikit-Learn, PyTorch, TensorFlow, Keras, Spark
Visualization/Other Tools: Tableau, Power BI, Jira, Microsoft Excel
Cloud Technologies: AWS (S3, EC2, ECR, SageMaker, Redshift, CloudWatch), Azure (Blob Storage, Virtual Machines, AI), GCP
Database/Software/Other Skills: MySQL, SQL Server, Oracle, MongoDB, Data Cleaning, Data Wrangling, Critical Thinking, Communication Skills, Presentation Skills, Problem-solving, Decision Making, EDA, Communication Skills, Databricks, Data Visualization, Predictive Analytics, Pattern Recognition, JMP, Data Integrity, Quantitative Data, Data Science, Statistics, Statistical Analysis, Data Analytics, Data Modeling, Big Query, Snowflake, Data Analysis, Data Mining, SAP, Mathematics, Computer Science, Programming, Time Series, Electronic Health Record (EHR), Hadoop
Operating System: Windows, Linux

WORK EXPERIENCE

Citigroup, USA | Data Scientist **March 2024 - Current**

- Facilitated Agile ceremonies, including sprint planning, daily stand-ups, sprint reviews, and retrospectives, ensuring seamless project execution and fostering a culture of continuous improvement in financial software development.
- Developed machine learning models leveraging statistical analysis and predictive modeling techniques to assess credit risk, fraud detection, and portfolio optimization, utilizing regression, classification, and clustering algorithms.
- Designed and implemented predictive analytics tools that segmented clients based on risk exposure, leading to a 20% improvement in default prediction accuracy and enabling proactive risk mitigation strategies for financial institutions.
- Built NLP-based models to automate financial document processing, fraud detection, and sentiment analysis for market trends, enhancing operational efficiency and investor sentiment insights.
- Developed and fine-tuned Large Language Models (LLMs) and Generative AI solutions for extracting key insights from unstructured financial reports, earnings calls, and regulatory filings, improving investment decision-making and compliance monitoring.
- Engineered scalable data pipelines using PySpark and cloud services to process massive financial datasets, integrating real-time analytics for market risk analysis, portfolio rebalancing, and algorithmic trading strategies.
- Applied deep learning techniques (CNNs, RNNs) to analyze handwritten financial documents, such as invoices, contracts, and loan applications, enabling automation in document processing and anomaly detection.
- Optimized marketing and investment strategies by tailoring campaigns to investor profiles, leading to a 20% increase in customer acquisition and a 30% boost in financial product conversions.
- Developed and deployed a web-based financial analytics platform utilizing Docker for containerization, AWS ECR for image storage, and EC2 for hosting. Streamlined deployment using CI/CD pipelines, managed source control with Git, and tracked experiments with MLflow to deliver real-time insights for risk management and investment teams.

- Integrated advanced Retrieval-Augmented Generation (RAG) systems with LLMs, LangChain, and Vector Databases, enabling financial analysts to access simplified reports for enhanced decision-making in lending, asset management, and regulatory compliance.
- Applied statistical validation techniques and SHAP analyses for financial risk assessments, reducing prediction errors by 10% and improving the accuracy of credit scoring models.
- Conducted A/B testing to optimize trading algorithms, customer engagement strategies, and risk models, leading to higher returns on investment and enhanced financial decision-making.
- Created interactive financial data visualizations using Tableau and Matplotlib, facilitating data-driven insights for portfolio managers, risk analysts, and CFOs, while collaborating with database administrators to optimize MySQL query performance for faster trade execution and financial reporting.

Nevina Infotech Pvt Ltd, India | Data Scientist **March 2019 - Aug 2022**

- Performed comprehensive testing and validation of data science models in accordance with Waterfall methodology, ensuring the models met stringent accuracy and reliability standards prior to deployment.
- Conducted A/B testing to assess the performance of marketing campaigns and product features, leading to actionable insights that improved overall campaign effectiveness and customer engagement.
- Optimized large-scale data processing workflows using Apache Spark, reducing processing times by 40% and enhancing the overall throughput and scalability of data handling operations.
- Utilized machine learning techniques such as Naïve Bayes, Logistic Regression, Random Forest, and XGBoost to predict loan default probabilities, resulting in improved risk management strategies with increased predictive accuracy.
- Enhanced model accuracy by 20% through the application of deep learning frameworks, including TensorFlow and PyTorch, effectively improving the performance of predictive models used for credit risk analysis.
- Automated complex data pipelines using Azure Data Factory and Azure Databricks, significantly increasing the efficiency of data engineering processes and minimizing manual intervention and potential errors.
- Developed real-time reporting solutions by integrating Power BI with various data sources, including Excel, to deliver timely, accurate insights that empowered business leaders to make data-driven decisions.
- Implemented robust backup, recovery, and disaster recovery solutions for SQL Server databases, ensuring data integrity, availability, and minimal downtime in case of failures or emergencies.

EDUCATION

Master’s in Data Science
University of Massachusetts Dartmouth, Dartmouth, MA

CERTIFICATIONS

MS Azure Machine Learning Scholarship by Udacity.
Google AI Explore ML Intermediate.
Data Science Tool box by Coursera.